Handheld SOP

Station 5

How to load weighed Packs into a cage one tag scan at a time, complete the cage, and what to do when Station 5 refuses a scan.

Version	Date	Device	Station
1.5.0	2026-11-06	Chainway C72	Station 5 cage loading

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01 Purpose and scope

The Station 5 app runs on the Chainway C72 handheld in the cage-loading area (Zone 8). Operators use it to **load weighed Packs into a cage** and to **complete the cage** when it is full. Every scan is checked by the Station 5 desktop, which keeps the cage's Pack list, count and weight.

What it does

- Arms a cage when you scan its RFID tag, then adds **one** Pack to it when you scan the Pack's QR label.
- Shows the cage's Pack count, total weight and the Packs already in it, as Station 5 records them.
- Completes the cage after you confirm a summary, and shows the SAP receipt and manifest print status.
- Stops you, full screen, when a Pack must not go into the cage.

What it does not do

- It does not weigh Packs or release them from QC. Those happen at the weigh station and on the desktop.
- It does not post to SAP or print the manifest. The Station 5 desktop does both after you complete the cage.
- It cannot take a Pack out of a cage. Corrections are made on the Station 5 desktop.

> **Who this is for**
> Cage-loading operators at Station 5. Sections marked **Supervisor** are for the shift supervisor, who holds the settings PIN.

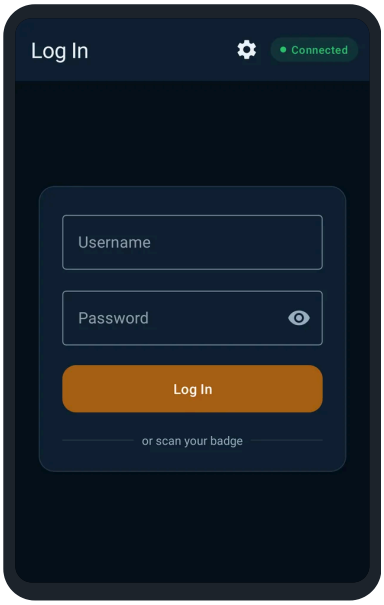
02 Before you start

- [ ] The handheld is charged. Put it back on its charger at the end of the shift.
- [ ] You are on the plant Wi-Fi, in coverage at the cages.
- [ ] You know your Station 5 username and password.
- [ ] The Station 5 desktop is running. The status pill shows **Connected** (section 3).
- [ ] Each cage has its RFID tag, and each Pack has its QR label.

Opening the app

The handheld starts in the **PPNAM Station Launcher**. Tap the amber **Station 5** tile.

The app opens on **Log In**, or straight on the cage screen if you are still signed in. Wait for the status pill in the top-right corner to settle on **Connected** before you start.



The Log In screen.

03 The status pill and header

The coloured pill in the top-right corner is on every screen. It reports two links: the **network link** (the broker that carries messages) and **Station 5 itself** (the desktop). It waits 1.5 seconds before it changes, so a short blip does not make it flicker.

Connected

The network is up and the Station 5 desktop is running. Everything works.

What to do: Carry on.

Reconnecting

The handheld lost the network link and is getting it back. It retries every 5 seconds.

What to do: Wait. If it does not clear, move into Wi-Fi coverage.

Station 5 offline

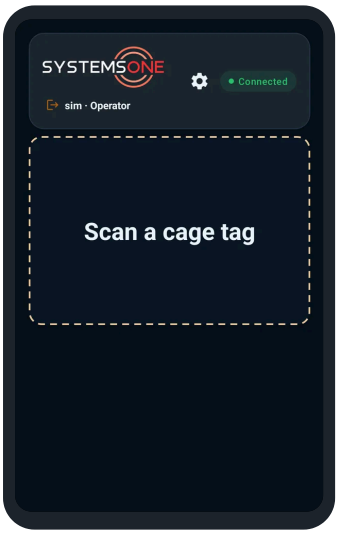
The network is fine, but the Station 5 desktop is not running or not reachable. The cage screen is covered by a warning.

What to do: Do not load Packs. Wait for **Connected** and tell your supervisor if it lasts.

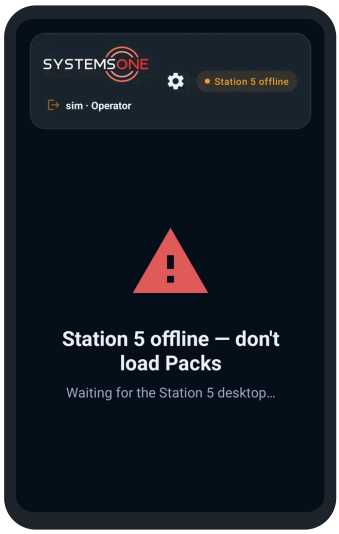
Offline

The handheld is not trying to connect: it has no broker account set up yet, or a supervisor is applying new settings.

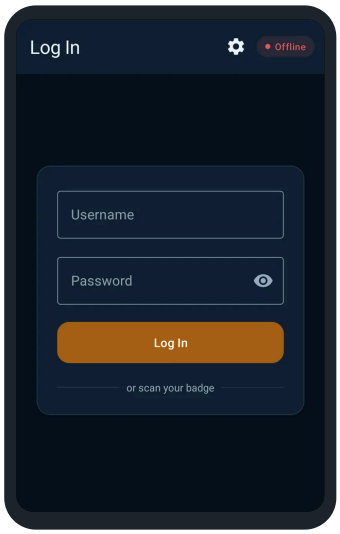
What to do: Ask your supervisor to set up the broker in Settings.



Connected: ready for a cage tag.



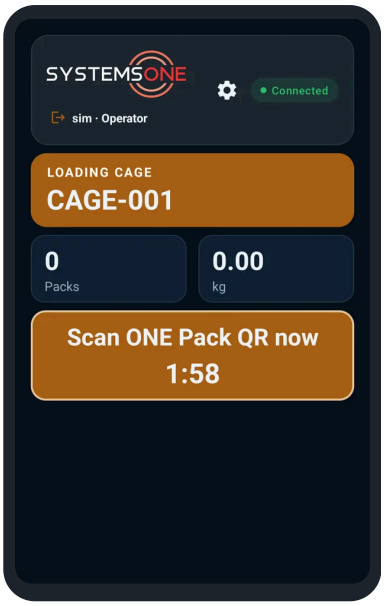
Station 5 offline: the cage screen is blocked.



Offline: no broker account set up yet.

The header

- **Your name and role** sit under the logo. Tap them to log out (section 10).
- The **gear** opens Settings.
- Below the header, the amber **LOADING CAGE** banner names the open cage and the job of its last Pack.



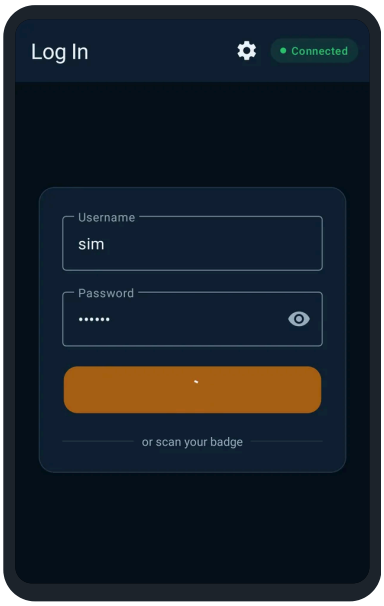
Your name and role, Settings, the pill, and the cage you are loading.

04 Signing in

Every Pack you load is recorded against your name, so the app does nothing until you sign in.

By username and password

1. Tap **Username** and type your username. Once the station has sent its list, an arrow lets you pick your name instead (each row reads *username — Your Name*).
2. Tap **Password** and type your password. The eye icon shows what you typed.
3. Tap **Log In**. A spinner replaces the button while Station 5 checks your password, and the fields are locked.

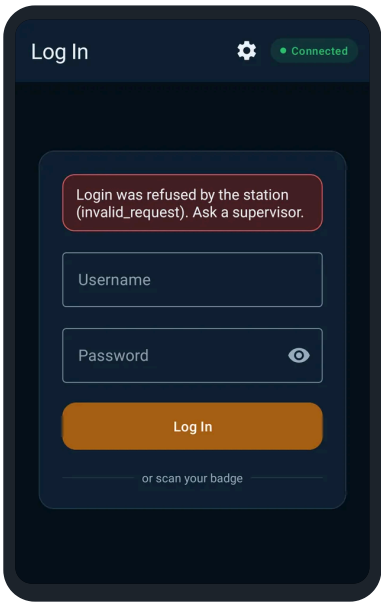


The spinner shows while Station 5 checks your password.

Badge sign-in is not available yet

The screen says *or scan your badge*, and a badge read is sent to Station 5, but the current Station 5 desktop does not accept badge sign-in. It answers *Login was refused by the station (invalid_request). Ask a supervisor.*

Sign in with your username and password.



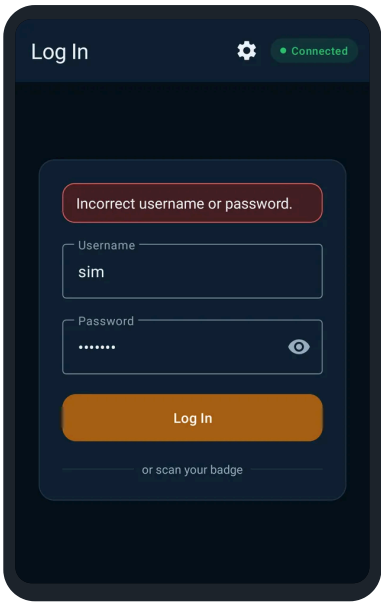
A badge read is refused by the current Station 5 desktop.

If sign-in fails

The reason appears in red at the top of the card. Typing in either box hides it again. The common ones:

- **Incorrect username or password.** Retype them carefully.
- **Station 5 did not respond. Check the station and retry.** No answer within 10 seconds. Check the pill.
- **Not connected to the broker. Check Settings and try again.** The handheld has no network link.
- **Please fill in all fields** Both boxes are needed.

Every sign-in message is listed in section 8.

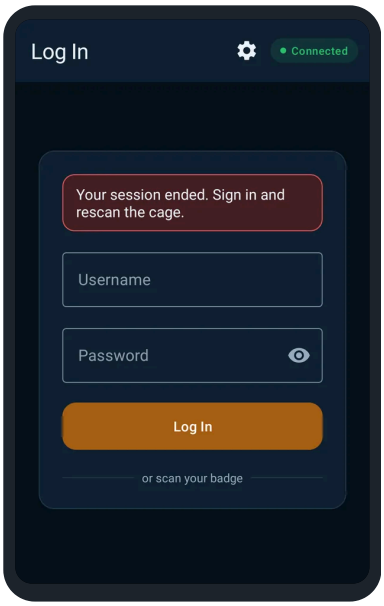


The reason appears in red above the fields.

Signed out by the app

- If nobody touches the handheld or scans for **15 minutes**, the app signs you out with *Signed out after 15 minutes of inactivity.* Your supervisor can change the time. Time with the app in the background counts too.
- If Station 5 ends your session during a scan, you see *Your session ended. Sign in and rescan the cage.*

Sign in again and scan the cage tag again.



Why you were signed out is shown on Log In.

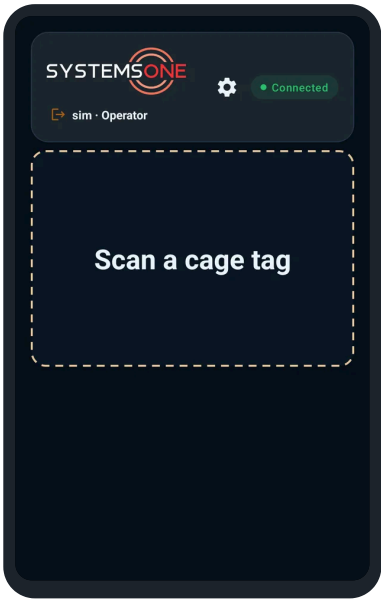
05 Loading a cage

The rule is **one tag scan arms one Pack**. Scan the cage's RFID tag, then scan **one** Pack's QR label within two minutes. Repeat for every Pack. Station 5 checks each Pack before it is added.

Step 1 Scan the cage tag

Hold the handheld to the cage's RFID tag and pull the trigger. The step box briefly reads **Sending…** while Station 5 answers.

If you scan a Pack first, the app stops you with *Do NOT load this Pack yet: Scan the cage tag first, then scan this Pack again.*

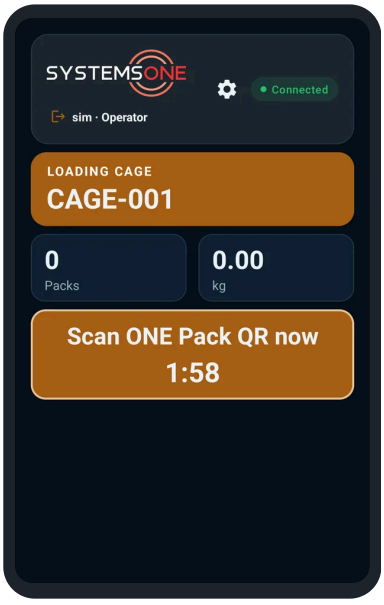


Nothing is armed yet.

Step 2 Scan ONE Pack QR now

When Station 5 accepts the tag you hear the success chirp. The cage banner turns amber and the box reads **Scan ONE Pack QR now** with a countdown from **2:00**.

Scan the QR label of the **one** Pack you are putting in the cage. The tiles show the cage's Pack count and total kg as Station 5 records them.

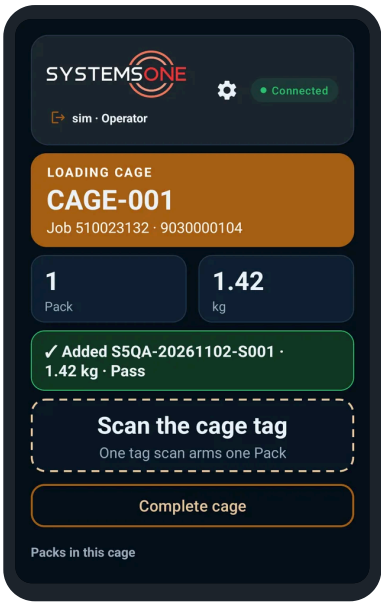


Armed for one Pack, with the time left.

Step 3 The Pack is added

You hear the success chirp and a green card reads, for example, *✓ Added S5QA-20261102-S001 · 1.42 kg · Pass*. **Pass** means it passed QC; **Manager accepted** means a manager released it.

Put the Pack in the cage. The arm is now spent: the box goes back to **Scan the cage tag** / *One tag scan arms one Pack*.

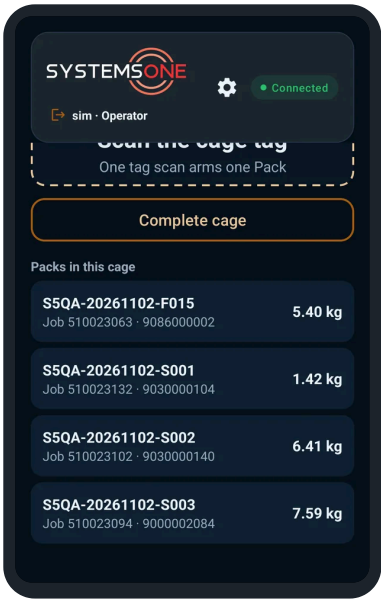


Added, and the cage is disarmed again.

Step 4 Repeat for every Pack

Scan the cage tag, then one Pack, for each Pack. Scroll down to see **Packs in this cage**: each row shows the Bag ID, *Job · outer code* and the weight.

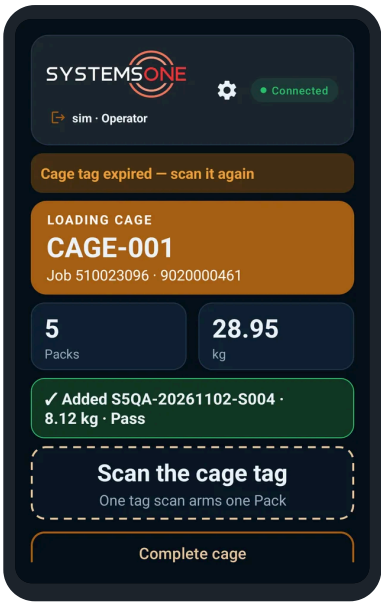
If you scan a Pack that is already in this cage, you hear the chirp and see *S5QA-… is already in this cage.* Nothing changes.



The Packs already in this cage, newest first.

If the two minutes run out

When the countdown reaches 0:00 the cage disarms and an amber notice reads *Cage tag expired — scan it again.* Nothing was sent. Scan the cage tag again, then the Pack.



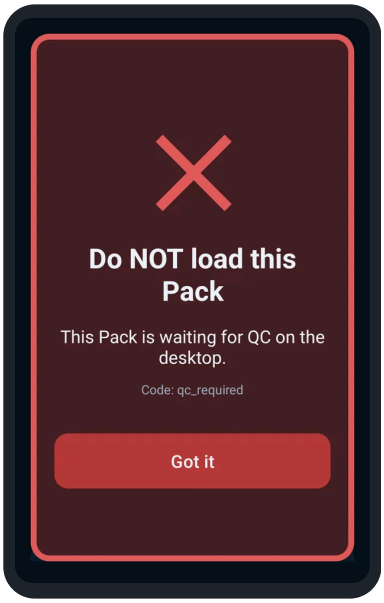
The arm expired before a Pack was scanned.

When a Pack is refused

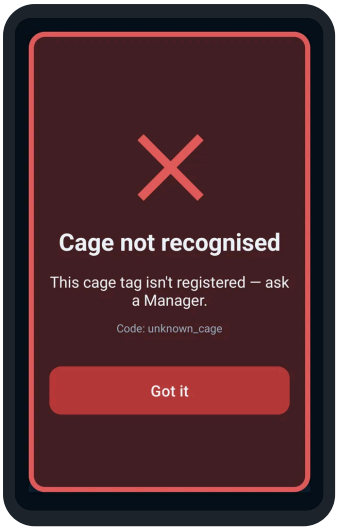
Do NOT load this Pack

A refusal fills the screen with a red ✕, a headline, what to do, and a code. You hear the fail buzz. **Do what the headline says.** For *Do NOT load this Pack*, keep that Pack out of the cage and put it aside.

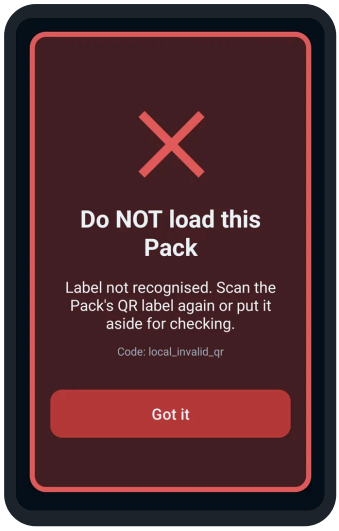
Tap **Got it** to carry on. Scans are blocked until you do. Section 8 lists every refusal.



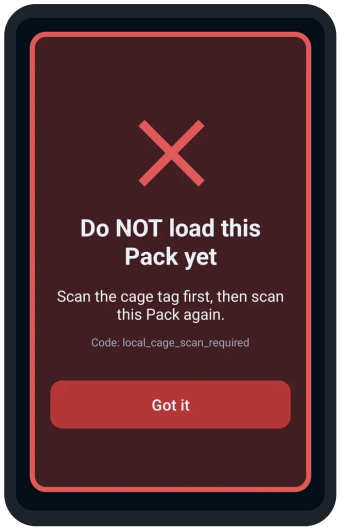
A refusal covers the whole screen until you tap Got it.



An unregistered cage tag. Ask a Manager.



A label that is not a Pack QR.



A Pack scanned before the cage tag.

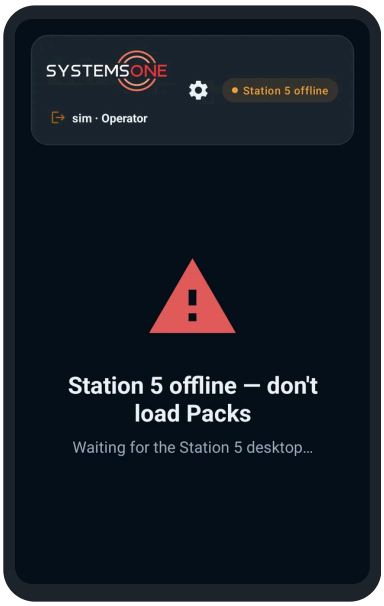
> **A bad label does not use up the arm**
>
> *Label not recognised* is decided on the handheld, so nothing is sent and the countdown carries on after **Got it**. Scan the right label before it runs out.

While Station 5 is busy or offline

Station 5 offline

If the desktop goes offline, the cage screen is covered by *Station 5 offline — don't load Packs* / *Waiting for the Station 5 desktop…*. Scans play the fail buzz and do nothing. The header stays usable, so you can still log out.

Wait for **Connected**. A scan that was already on its way is kept and sent again until Station 5 answers.

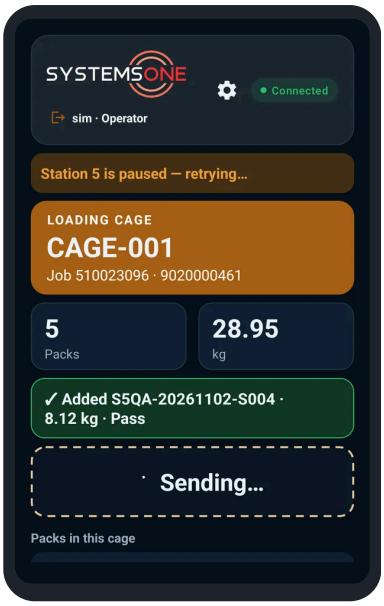


Do not load Packs while this shows.

Paused or busy

Station 5 is paused — retrying… or *Station 5 is busy — retrying…* means the desktop has your scan but cannot act on it yet. The app resends it every 10 seconds; wait for the answer.

While a scan shows **Sending…**, a scan of a **different** code plays the fail buzz with *Wait for Sending… to finish, then scan again.* A repeat read of the same code is ignored.



The station is paused. The app keeps retrying.

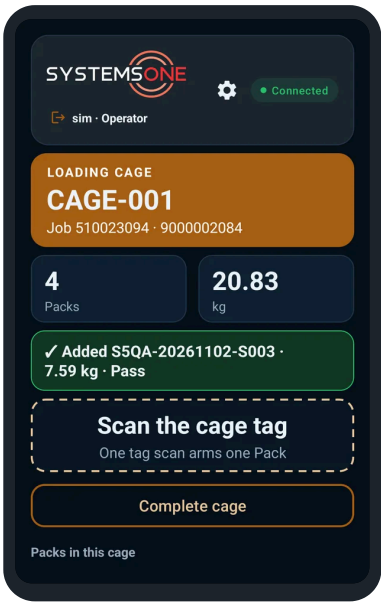
> ⚠ **After a pause, rescan the cage**
>
> While Station 5 is paused, a Pack scan still uses up the arm. When it resumes you may see *Scan the cage tag first*: scan the cage tag again, then the Pack.

06 Completing a cage

When the cage is full, complete it. After that no more Packs can be added, and the Station 5 desktop posts the SAP receipt and prints the manifest.

Step 1 Tap Complete cage

Complete cage sits under the step box. It shows only when the cage is open, holds at least one Pack, and nothing is sending. Tap it.



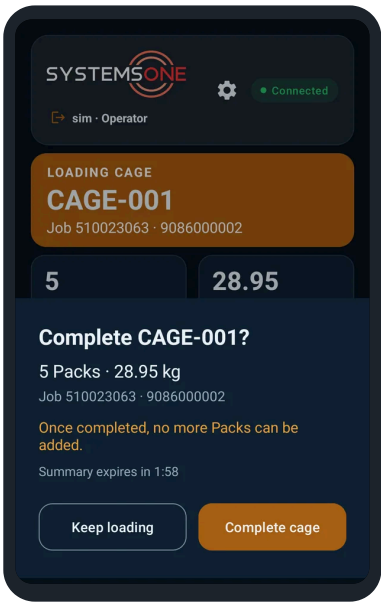
Complete cage appears once the cage holds a Pack.

Step 2 Check the summary

A sheet asks *Complete CAGE-001?* and shows the Pack count, total weight and a job line, with *Once completed, no more Packs can be added.*

- Check the count and weight against the cage.
- Tap **Complete cage** to confirm, or **Keep loading** to go back without changing anything.
- The summary is valid for two minutes (*Summary expires in 1:58*). If it runs out you see *Summary expired — tap Complete cage again.*

Scans are ignored while the sheet is open.



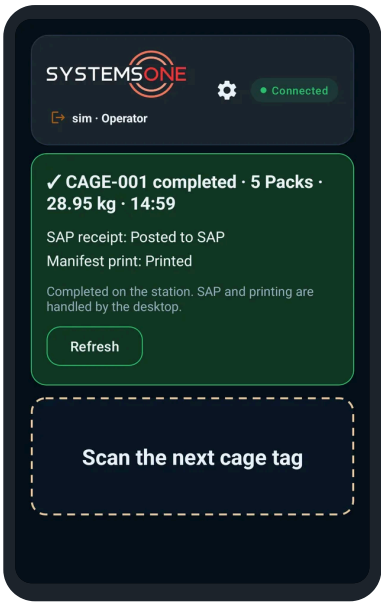
The summary to check before completing.

> **The job line shows one job**
> If the cage holds Packs from more than one job, the summary's job line still shows a single job. Check the count and weight, not the job line.

Step 3 The cage is completed

A green card confirms it, for example *✓ CAGE-001 completed · 5 Packs · 28.95 kg · 14:59*, with *Completed on the station. SAP and printing are handled by the desktop.* Completing makes no sound.

SAP receipt and **Manifest print** show how the desktop is getting on. Tap **Refresh** to update them. Amber text means the desktop needs attention: tell your supervisor. Then scan the next cage tag.



Completed, posted to SAP and printed.

SAP receipt and manifest print status

Status	What it means
Not sent yet / Not printed yet	The desktop has not started.
Pending	Waiting in the desktop's queue.
Sending to SAP / Printing	In progress.
Waiting for receipt	The manifest prints after SAP confirms the receipt.
Posted to SAP / Printed	Done.
Failed — check the desktop	It failed. Tell your supervisor.
Unconfirmed — check the desktop	The desktop could not confirm it. Tell your supervisor.
Needs review on the desktop	A person must check it on the desktop.

07 Scan sounds and vibration

Every PPNAM handheld uses the same two sounds. The Chainway scanner beeps by itself when it reads anything; the app's sound comes **after** that, when Station 5 has answered.

- **Success chirp** (a bright rising two-note chirp) with one short 40 ms vibration: the scan was **accepted**.
- **Fail buzz** (a low falling double buzz) with two 200 ms vibrations: the scan **failed**. Read the screen.

The sounds play at the **media volume**. If you cannot hear them, turn the media volume up; the vibration still tells you.

What you hear in the Station 5 app

When	Outcome	Sound
Cage tag	Station 5 accepts the cage and it is open for loading	● Success chirp
Cage tag	Refused (unknown cage, cage not available), or accepted with no open load	● Fail buzz
Pack QR	Added to the cage, or already in this cage	● Success chirp
Pack QR	Refused by Station 5 (*Do NOT load this Pack* and others)	● Fail buzz
Pack QR	Scanned before the cage tag, or the label is not a Pack QR	● Fail buzz
Any scan while *Sending…*	A different code	● Fail buzz
Any scan while *Sending…*	The same code again (a repeat read)	○ Silent
Any scan while *Station 5 offline* shows	Refused on the handheld	● Fail buzz
Any scan while a refusal screen or the summary sheet shows	Ignored	○ Silent
Station 5 is paused / busy — retrying…	Waiting for the final answer	○ Silent
Your session ended during a scan	Signed out	● Fail buzz
Complete cage, Keep loading, Refresh, Got it	Buttons	○ Silent
Log In, Settings	Signing in, settings	○ Silent

The repeat-read rule

One pull of the trigger can read the same tag several times. A repeat of the code that is **already sending**, or of a code **just refused on the handheld** within 1.5 seconds, makes no sound. A different code plays the fail buzz once. When Station 5 answers, you hear exactly one sound for that scan.

08 Messages and what to do

Every message the app shows, word for word. *Ask a Manager* or *Ask a supervisor* means the problem cannot be fixed on the handheld.

Signing in

Message	What it means	What to do
Please fill in all fields	The username or password box is empty.	Fill in both.
Incorrect username or password.	Station 5 did not accept the password.	Retype carefully. Ask your supervisor if you are locked out.
Station 5 did not respond. Check the station and retry.	No answer within 10 seconds.	Check that the desktop is running, then retry.
Not connected to the broker. Check Settings and try again.	The handheld has no network link.	Check the pill and Wi-Fi, then retry.
Login was refused by the station (*code*). Ask a supervisor.	Station 5 refused for another reason, for example `processing_paused`, or `invalid_request` for a badge.	Use your username and password. Tell your supervisor the code if it repeats.
Login failed. Try again.	Sign-in failed without a reason.	Try again.
Station sent an incomplete / invalid / unusable authentication challenge	The station's answer could not be used.	Retry. Report it if it repeats.
Station's authentication challenge did not match this device's request	The answer was for another request.	Retry.
Station failed authentication verification — this response is not trusted	The answer could not be verified.	Do not keep retrying. Tell your supervisor.
Station accepted the login but issued no session / Station closed this session immediately	The desktop did not open a session.	Retry, then tell your supervisor.
Signed out after *N* minutes of inactivity.	The automatic sign-out ran.	Sign in again.
Your session ended. Sign in and rescan the cage.	Station 5 ended your session during a scan.	Sign in, then scan the cage tag again.

Refusal screens (tap Got it after doing what it says)

Headline and message	Code	What to do
Do NOT load this Pack yet: Scan the cage tag first, then scan this Pack again.	`local_cage_scan_required`	Scan the cage tag, then the Pack.
Do NOT load this Pack: Label not recognised. Scan the Pack's QR label again or put it aside for checking.	`local_invalid_qr`	Scan the QR label, not another barcode. Put a damaged Pack aside.

Headline and message	Code	What to do
Do NOT load this Pack: *Bag ID* is already in another cage. Put it aside and check it on the desktop.	`pack_allocated`	Put it aside; a supervisor checks the desktop.
Do NOT load this Pack: This Pack hasn't been weighed. Take it back to the weigh station.	`pack_not_found`	Take it to the weigh station.
Do NOT load this Pack: This Pack is waiting for QC on the desktop.	`qc_required`	Put it aside until QC releases it.
Do NOT load this Pack: Label doesn't match the weighed Pack. Put it aside for checking.	`invalid_barcode`, `outer_code_changed`	Put it aside for checking.
Do NOT load this Pack: Job card on this label doesn't match. Put it aside for checking.	`production_order_…`	Put it aside for checking.
Do NOT load this Pack: This Pack's outer code doesn't match the cage. Use a matching cage.	`mixed_outer_code`	Load it into a cage for that outer code.
Scan the cage tag first: Scan the cage tag again, then scan this Pack within two minutes.	`cage_scan_required`	Scan the cage tag, then the Pack.
Cage not recognised: This cage tag isn't registered — ask a Manager.	`unknown_cage`	Use another cage and tell a Manager.
Cage not available: This cage is already completed.	`cage_locked`	Use another cage.
Cage not available: This cage is no longer active on this handheld. Rescan the cage tag.	`cage_not_active`, `cage_context_changed`	Scan the cage tag again.
Cannot complete: Add Packs before completing.	`empty_cage`, `completion_blocked`	Load at least one Pack first.
Cage not completed: The summary expired. Tap Complete cage again.	`confirmation_expired`	Tap **Complete cage** again.
Cage not completed: Another Pack was added to this cage. Check the new count and tap Complete cage again.	`cage_contents_changed`	Check the new count, then complete again.
Check the handheld's clock: The handheld's time is wrong. Ask a supervisor to correct it, then scan again.	`request_expired`	Ask a supervisor to set the clock.
Not accepted: This request clashed with an earlier one. Note the time and contact support. Ref: …	`message_id_conflict`	Note the time and the Ref; contact support.
Not accepted (or Cage not completed): *the station's reason*	any other code	Read the reason; tell a Manager.

Notices above the cage

Message	What it means	What to do
Cage tag expired — scan it again	Two minutes passed without a Pack.	Scan the cage tag again.
***Bag ID* is already in this cage.**	The Pack was already loaded here.	Nothing; carry on.
Wait for Sending… to finish, then scan again.	You scanned another code while one was sending.	Wait, then scan again.
Station 5 is paused — retrying… / Station 5 is busy — retrying…	The desktop cannot act yet.	Wait; the app retries every 10 seconds.
Summary expired — tap Complete cage again	The completion summary timed out.	Tap **Complete cage** again.
Tap Complete cage again to see the new count.	The cage changed before you confirmed.	Tap **Complete cage** again.
Bag ID* was recorded in *cage	After a new sign-in: your last Pack did get in.	Nothing.
Last Pack *Bag ID* was not confirmed. Rescan cage *cage*, then rescan that Pack.	After a new sign-in: the last Pack's outcome is unknown.	Rescan the cage, then that Pack.
Cage *cage* is still open. Rescan the cage tag to review.	After a new sign-in: your cage is still open.	Scan the cage tag.
Couldn't confirm the last scan. Rescan the cage tag.	After a new sign-in.	Scan the cage tag.

Settings

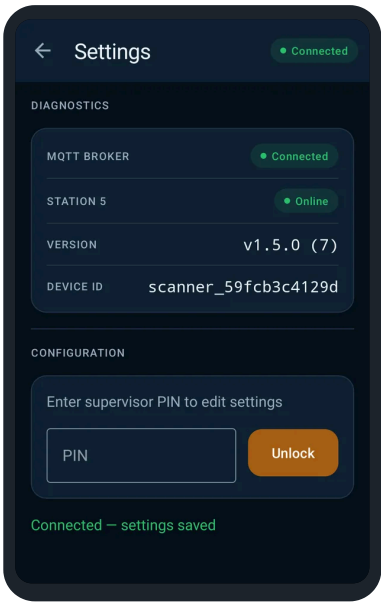
Message	What it means	What to do
Incorrect PIN. *N* attempts left before lockout.	Wrong PIN.	Supervisor only.
Too many attempts. Try again in *N* s.	Five wrong PINs lock the box for 30 seconds.	Wait for the countdown.
Enter the broker host / Enter a port from 1 to 65535 / Enter the broker username / Enter the broker password / Enter 0–1440	A field is missing or out of range.	Supervisor: fill in the field.
Could not store the password securely	The handheld could not save the password.	Try again; report it if it repeats.
Saved, but could not connect to *host*:*port*. Check the broker settings.	The settings were saved but do not connect.	Supervisor: check the values.

09 Settings

The gear opens Settings from Log In or the cage screen. The top half is open to everyone and is what support will ask you to read out. The bottom half needs the supervisor PIN.

Diagnostics: anyone

- **MQTT Broker**: the network link on its own (Connected, Reconnecting, Offline).
- **Station 5**: the desktop (Online, Offline, Unknown). Broker *Connected* with Station 5 *Offline* means the network is fine and the desktop is the problem.
- **Version**: `v1.5.0 (7)` for this release.
- **Device ID**: this handheld's identity, such as `scanner_59fcb3c4129d`. Read it out when you report a fault.
- **Signed in as** and **Log out** appear at the bottom when you opened Settings while signed in.



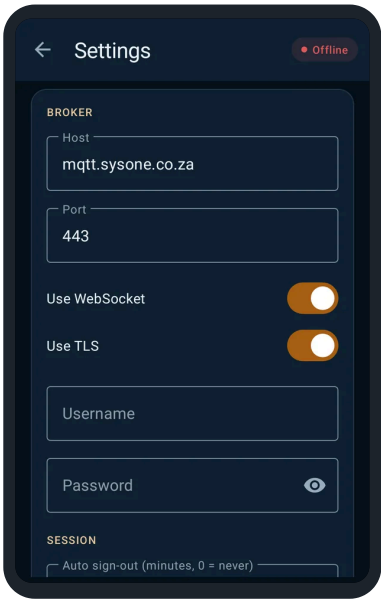
Diagnostics, and the PIN gate below.

> **Configuration**
> Type the supervisor PIN and tap **Unlock**. Five wrong attempts lock the PIN box for 30 seconds, and the count survives leaving Settings. A supervisor's settings tag also opens Settings from any screen. Never tell an operator the PIN; operators should ask the supervisor.

Unlocked: supervisors only

- **Host** and **Port**: normally `mqtt.sysone.co.za` and `443`.
- **Use WebSocket** and **Use TLS**: both on for the standard setup.
- **Username** and **Password**: the broker account, not an operator account. A new handheld has none, so it shows **Offline** until they are filled in. The password box is always empty; leave it blank to keep the saved one.
- **Auto sign-out (minutes, 0 = never)**: 15 by default, up to 1440.

Test & Apply saves the settings, shows *Testing connection…* and reconnects. *Connected — settings saved* means it worked, and the section locks again after two seconds. Otherwise it says *Saved, but could not connect to host:port. Check the broker settings.*



Unlocked, showing the standard broker settings on a new handheld.

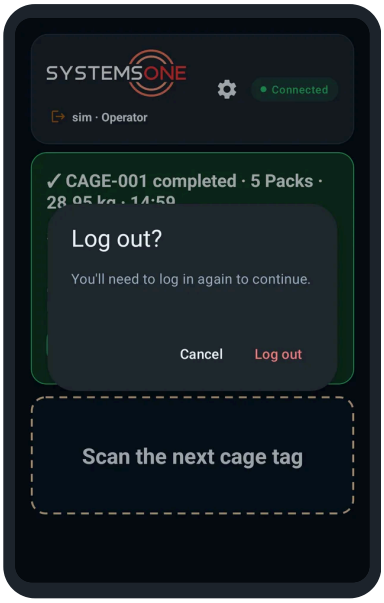
10 Finishing up

Handing the handheld over

1. Make sure nothing shows **Sending…**.
2. Tap your name in the header (or **Log out** in Settings).
3. Confirm **Log out**.

If a scan is still being sent, the dialog asks *A scan is still being sent. Sign out anyway?* Tap **Cancel** and wait for the answer. If you sign out anyway, the app checks that scan at the next sign-in and tells you what to rescan.

Always log out before you pass the handheld to someone else. Otherwise their Packs are recorded under your name.

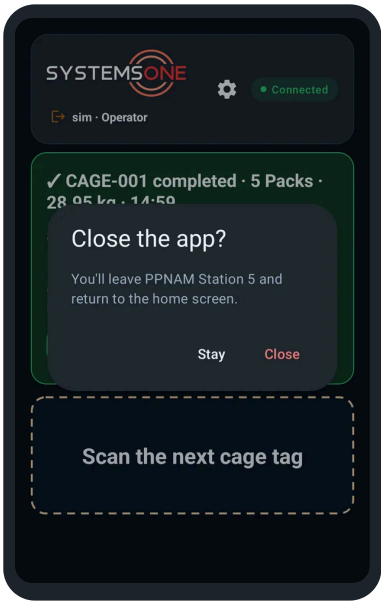


Log out asks first.

Closing the app

From Log In or the cage screen, the **Back** key asks *Close the app?*. Tap **Close** to return to the Launcher, or **Stay**.

At the end of the shift, log out and put the handheld on its charger.



Back from Log In or the cage screen.

11 Troubleshooting

Common problems

Problem	Likely cause	What to do
Station 5 offline — don't load Packs **covers the screen.**	The desktop is off or its program is not running.	Do not load Packs. Wait for **Connected**; tell your supervisor.
The pill stays Reconnecting.	No Wi-Fi where you are, or wrong broker settings.	Move into coverage. If it stays, your supervisor checks Settings.
The pill is red Offline on a new handheld.	No broker account is set up.	Your supervisor fills in Settings.
Every Pack says ***Scan the cage tag first.***	The arm expired, was spent, or a pause used it up.	Scan the cage tag immediately before each Pack.
Sending… does not clear.	The answer was lost. The app resends every 10 seconds.	Wait. If it lasts, check the pill.
Complete cage is missing.	The cage has no Packs, or a scan is still sending.	Load a Pack, or wait for Sending… to clear.
Failed — check the desktop **after completing.**	SAP or the printer failed on the desktop.	Tell your supervisor. The cage is still completed.
No scan sounds.	Media volume is down.	Turn up the media volume. The vibration still works.
A badge does not sign you in.	Station 5 does not accept badge sign-in.	Use your username and password.
Signed out unexpectedly.	15 minutes without a touch or scan, or Station 5 ended the session.	Sign in again and rescan the cage tag.

> **Reporting a fault**
> Give your supervisor the **Device ID** and **Version** from Settings, the time, the cage and Bag ID, and the exact message and code on screen.

12 Revision history

Version	Date	Change
`1.5.0`	2026-11-06	Suite release: visual refresh, consistency pass, scan sounds. First Station 5 handheld SOP.